

		Since the spring is in equilibrium, $F = kx = mg$ Hence gain in EPE = $\frac{1}{2} (kx)x = \frac{1}{2} mgx = \frac{1}{2} \text{ GPE}$
9	D	Maximum tension experienced at bottom of the swing. Centripetal force = tension – weight Using conservation of energy, kinetic energy at the bottom = loss of GPE $\frac{1}{2}mv^2 = mgr$ $\text{Centripetal force} = m \frac{v^2}{r} = \frac{2mgr}{r} = 2mg$ $2mg = \text{tension} - mg \rightarrow \text{tension} = 3mg$
10	D	$F = -\frac{dU}{dr}$, hence option D is correct while option A and B are incorrect.